

Introduction

Traveling waves are observed often throughout the brain. Their function is a matter of continuous debate, but evidence suggests that they may be connected to learning and information processing [1]. Because the shape and direction of traveling waves are linked to relevant processes, preservation of these properties may be an important mechanism of memory and learning. To investigate this, we have constructed a computational model network capable of propagating neuronal traveling waves with an included mechanism of synaptic plasticity (adjustment of the connection strength between neurons).

Methods

The basic unit of information at the level of the brain is thought to be the neuronal spike, in which the membrane potential of a neuron quickly depolarizes, followed by a fast hyperpolarization. When a neuron spikes, it transmits an electrical signal down it's axons (connections) to all downstream neurons. To simulate this, we have chosen the izhikevich model neuron [2], with individual neurons spatially placed into a 100 X 100 X 3 lattice. Connections, or synapses, between neurons are chosen randomly, with the probability of connection decaying exponentially with distance.

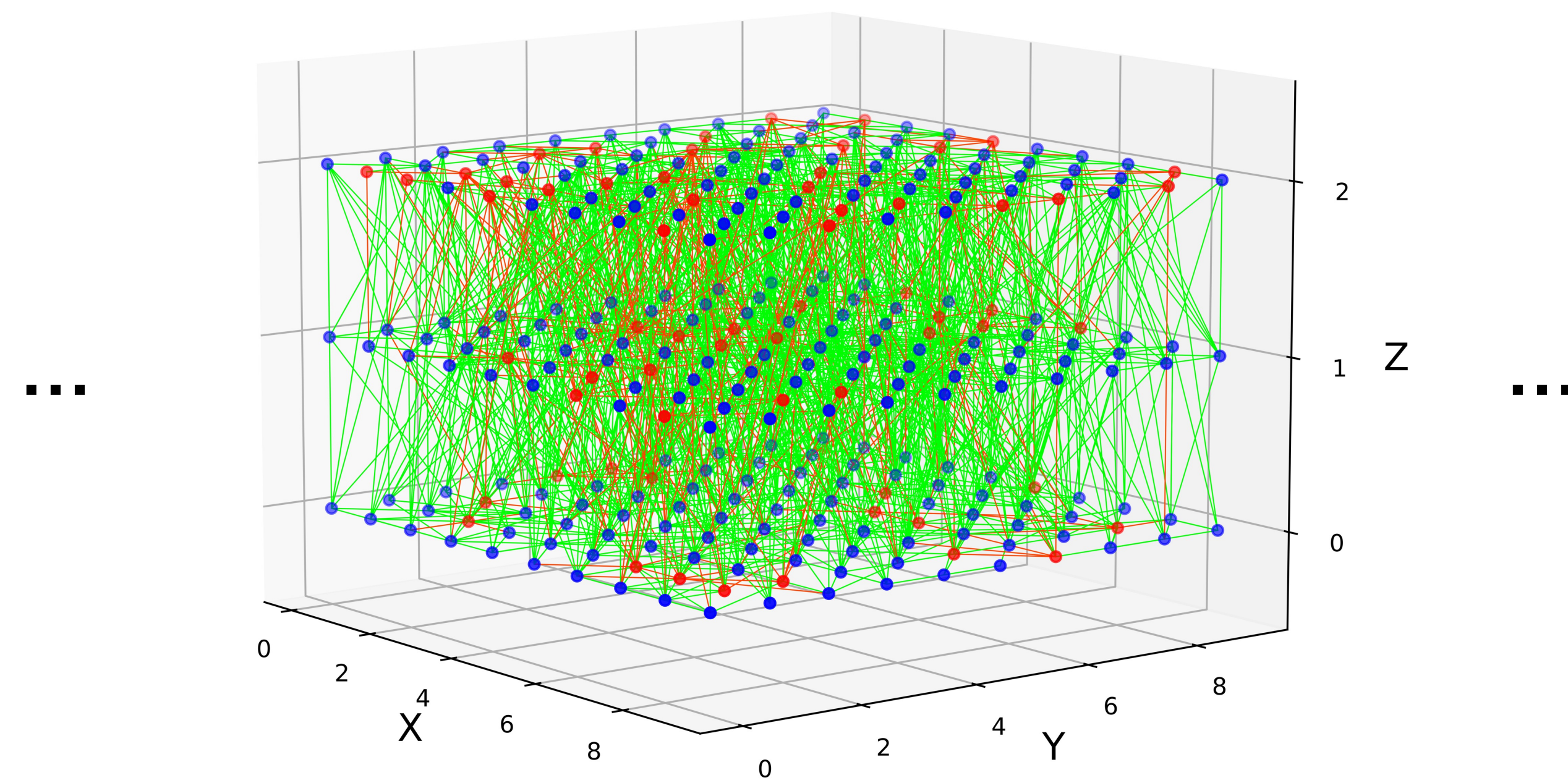


Figure 1: A 10 X 10 X 3 visualization of our network structure. Blue dots represent excitatory neurons (that transmit positive signals) while red dots represent inhibitory neurons (that transmit negative signals). Green connections originate from excitatory neurons while red/orange connections likewise originate from inhibitory neurons. NOTE: actual network dimensionality is 100 X 100 X 3.

The arrival of a spikes along an axon is delayed dependent on the distance between the connected neurons. The magnitude of input is chosen randomly, labeled the weight, W_{ij} . This forms the basis of information transfer within our network and is the mechanism of traveling wave propagation. To stimulate our network, we use input spikes from outside the network, either in specific locations or randomly across the network for less predictable wave formations.

$$\Delta W_{ij}^{exc} = \begin{cases} +R(a_+)e^{-\frac{|\Delta t|}{\tau_+}} & \text{if } \Delta t < 0 \\ -R(a_-)e^{-\frac{|\Delta t|}{\tau_-}} & \text{if } \Delta t > 0 \end{cases}$$

$$\Delta t = t_i - t_j$$

The above equation includes the implemented mechanism of synaptic adjustment, known as spike-timing dependent plasticity (STDP) [3], for which excitatory weights change dependent on the relative spike timing of connected neurons. Simulations are performed using the BRIAN neural simulator [4].

Results

Visualizations of Network Activity

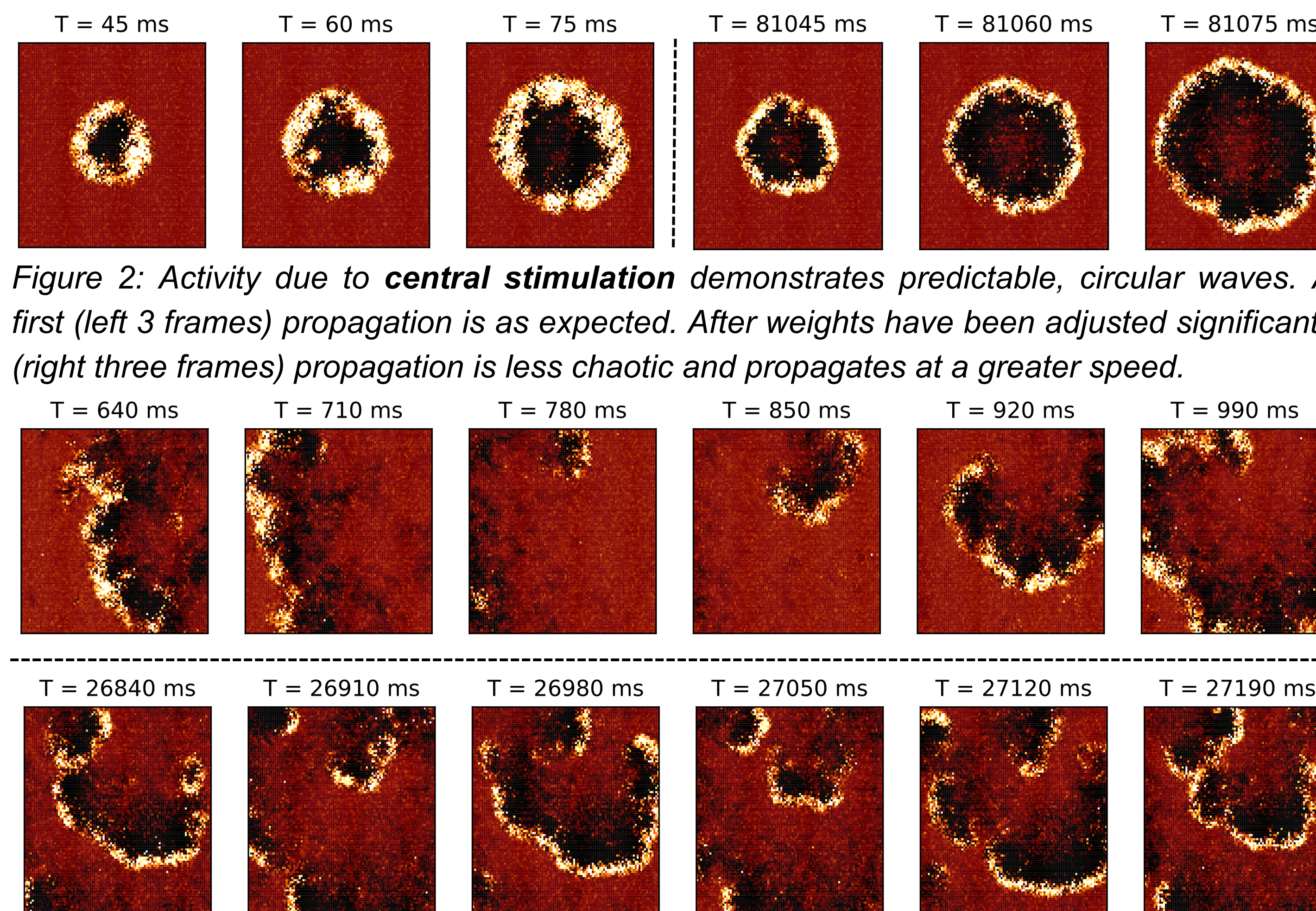


Figure 2: Activity due to **central stimulation** demonstrates predictable, circular waves. At first (left 3 frames) propagation is as expected. After weights have been adjusted significantly (right three frames) propagation is less chaotic and propagates at a greater speed.

Figure 3: Activity due to **stochastic stimulation**. At first (top row), waves propagate chaotically, beginning at any location within the network. Eventually (bottom row), pathways are formed and repetitive wave patterns emerge.

Pathway Formation (Central Stimulation)

Pathway formation can be defined as an increase in the spatial consistency of propagation for neural signals. Figures 4 - 7 demonstrate this in our networks for both central stimulation as well as for more chaotic inputs (stochastic stimulation)

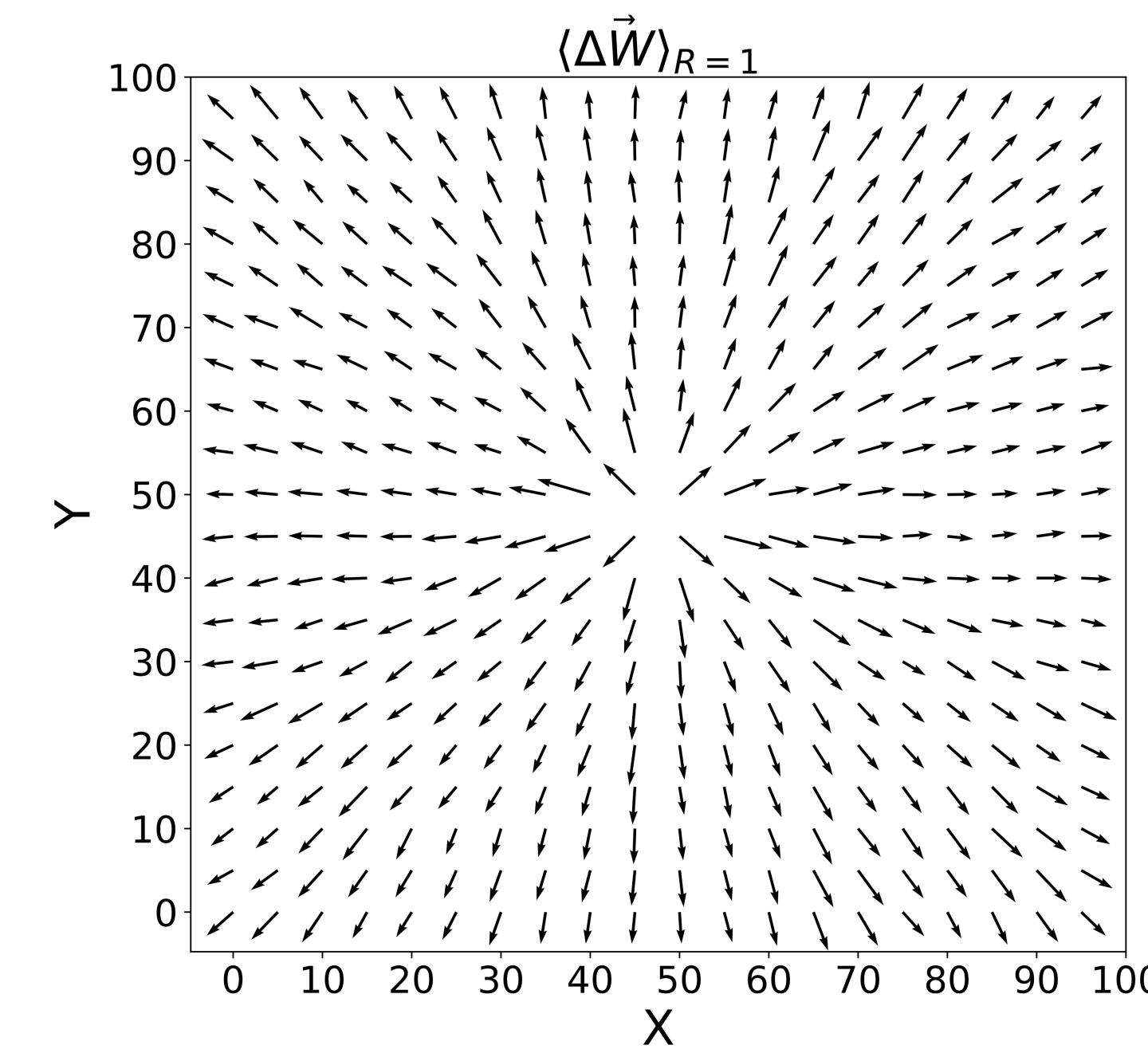


Figure 4: Vectors representing the mean changes in weight for every 5 X 5 region of the network, averaged across 40 random seeds. In general, with repetitive stimulation into the center of the network, outward wave propagation creates outward pathways when paired with STDP.

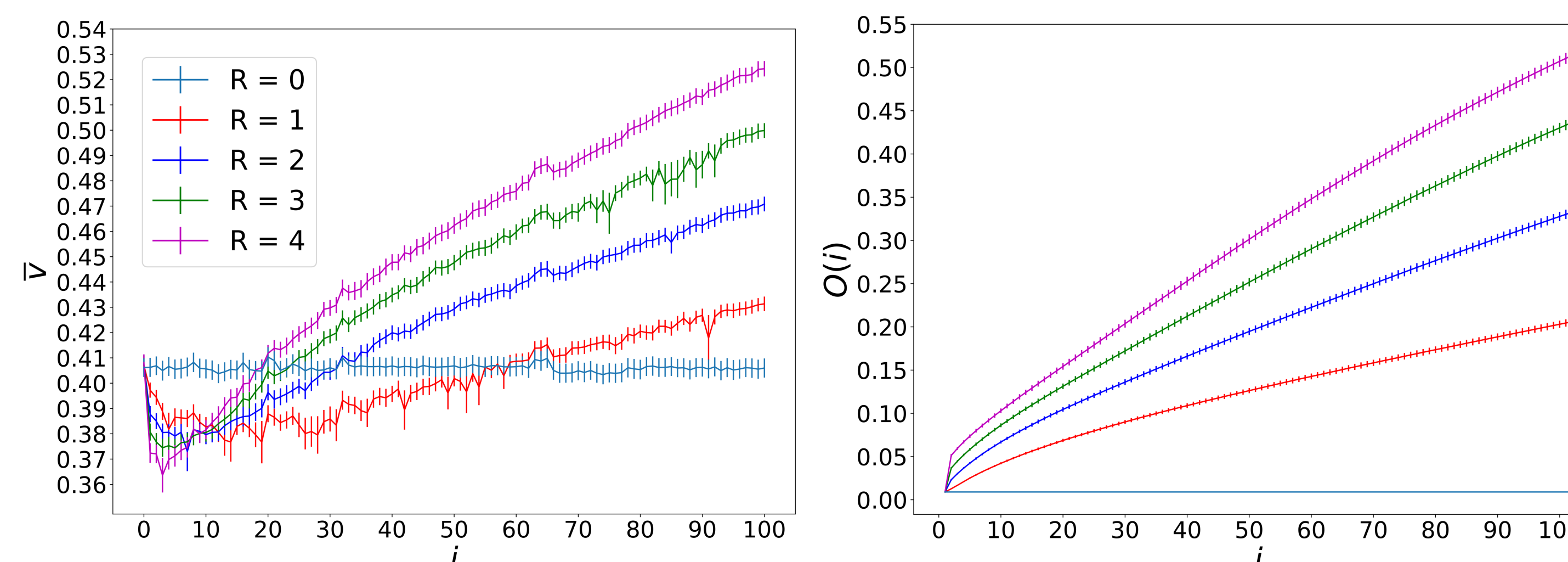


Figure 5: Averaged data for central stimulation. With repeated outward propagation, outward propagation speed (Left) increases over time. Additionally, our measure of pathway formation, the local order of the average outgoing weight for adjacent neurons (right) increases, suggesting the formation of pathways with each successive stimulation. NOTE: R is a variable we used to systematically adjust the scale of individual weight changes.

Pathway Formation (Stochastic Stimulation)

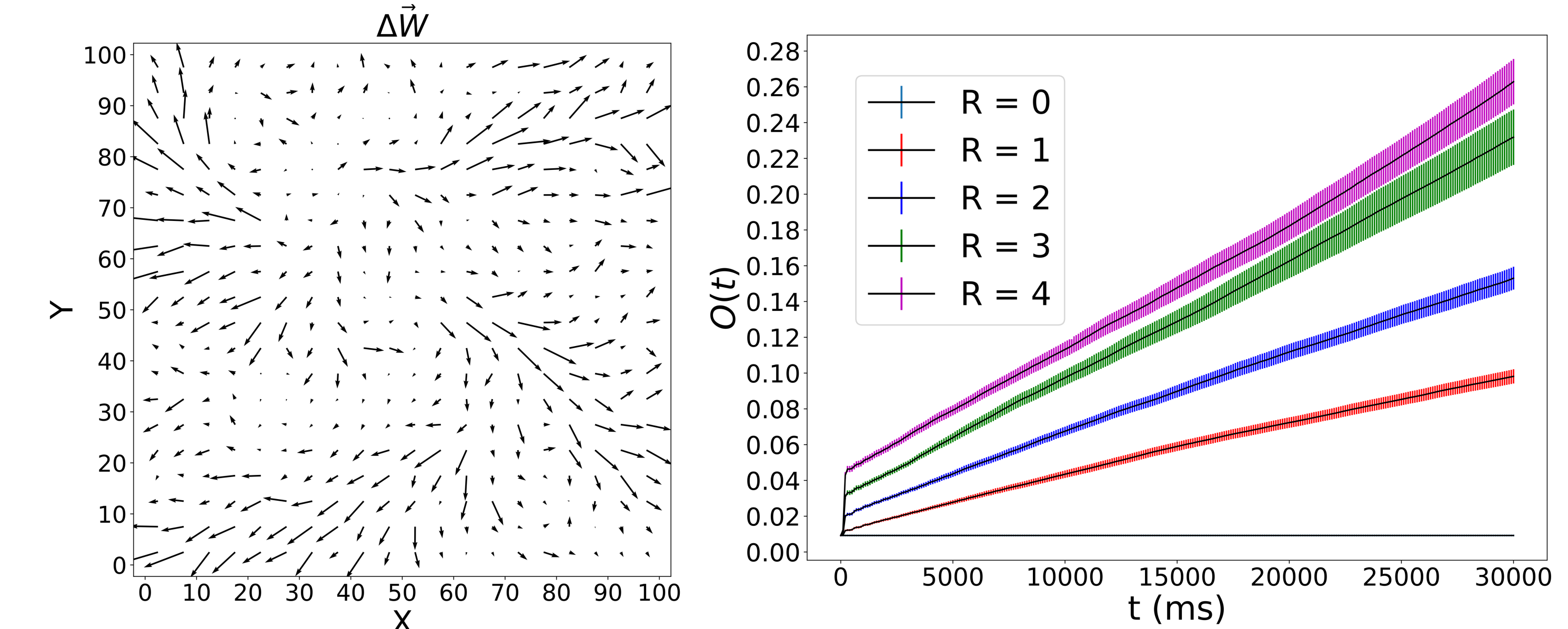


Figure 6: Even for random stimulation, the average changes in weight (left) for a single random seed still demonstrate the formation of spatial pathways. The measure of our local order parameter, $O(t)$ (right), also demonstrates consistent increases, demonstrating that even with stochastic stimulation, pathways still form over time.

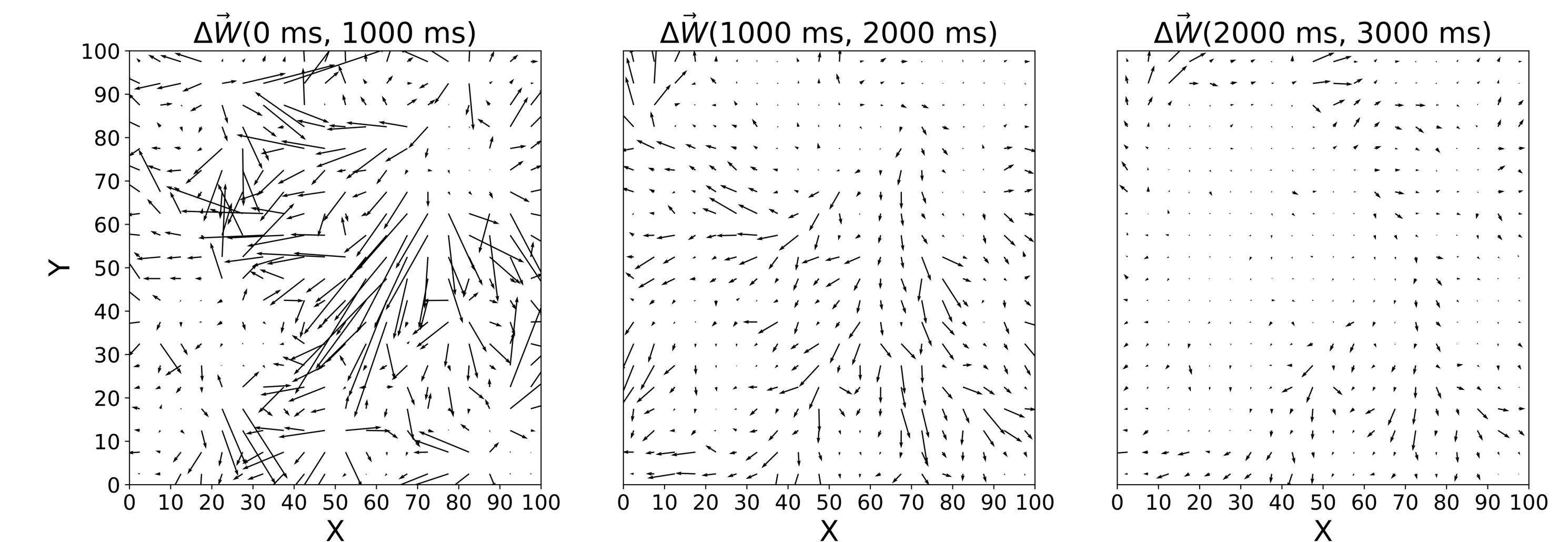


Figure 7: Vector maps of average changes in weight for every 5 X 5 region throughout the network for the first three 1000 ms time periods for a single case with random stimulation. The most significant changes in weight for an initially random network occur early, with more gradual synaptic adjustments occurring later in simulation.

Discussion and Conclusions

- **Traveling waves cause synaptic weights to adjust in an ordered and predictable way when paired with STDP.**
- The formation or strengthening of pathways due to these waves could be a mechanism of learning in regions where they are observed.
- If traveling waves are responsible for the strengthening of specific pathways, they may also be associated with an increase in propagation speed
- Understanding the effect that observed traveling waves have on biological neuronal networks of neurons is an important step towards uncovering the function(s) of this phenomenon.

References

- [1] Muller, Lyle, Frédéric Chavane, John Reynolds, and Terrence J. Sejnowski. 2018. "Cortical Travelling Waves: Mechanisms and Computational Principles." *Nature Reviews. Neuroscience* 19 (5): 255–68.
- [2] Izhikevich, E. M. 2003. "Simple Model of Spiking Neurons." *IEEE Transactions on Neural Networks / a Publication of the IEEE Neural Networks Council* 14 (6): 1569–72.
- [3] Song, Sen, Kenneth D. Miller, and L. F. Abbott. 2000. "Competitive Hebbian Learning through Spike-Timing-Dependent Synaptic Plasticity." *Nature Neuroscience*. <https://doi.org/10.1038/78829>.
- [4] Stimberg, Marcel, Romain Brette, and Dan Fm Goodman. 2019. "Brian 2, an Intuitive and Efficient Neural Simulator." *eLife* 8 (August). <https://doi.org/10.7554/eLife.47314>.